UNIT 4 Linear Momentum

4.1 Linear Momentum

4.2 Change in Momentum and Impulse

4.3 Conservation of Linear Momentum

4.4 Elastic and Inelastic Collisions

4.1 Linear Momentum

- **When two bodies collide, the effect they have on each other depends on both their initial velocities and mass.**
- **The momentum of a moving object = mass x velocity**
- **So momentum has a size and a direction (it's a vector quantity).**
- **The unit for momentum is the kilogram meter/second (kg*m/s)**
- **The symbol for momentum is *p*.**
- **For an object of mass m moving at a velocity v, its momentum**

$$p = m \times v$$

Momentum

Momentum ≡ Mass · Velocity

$$\vec{p} = m\vec{v}$$

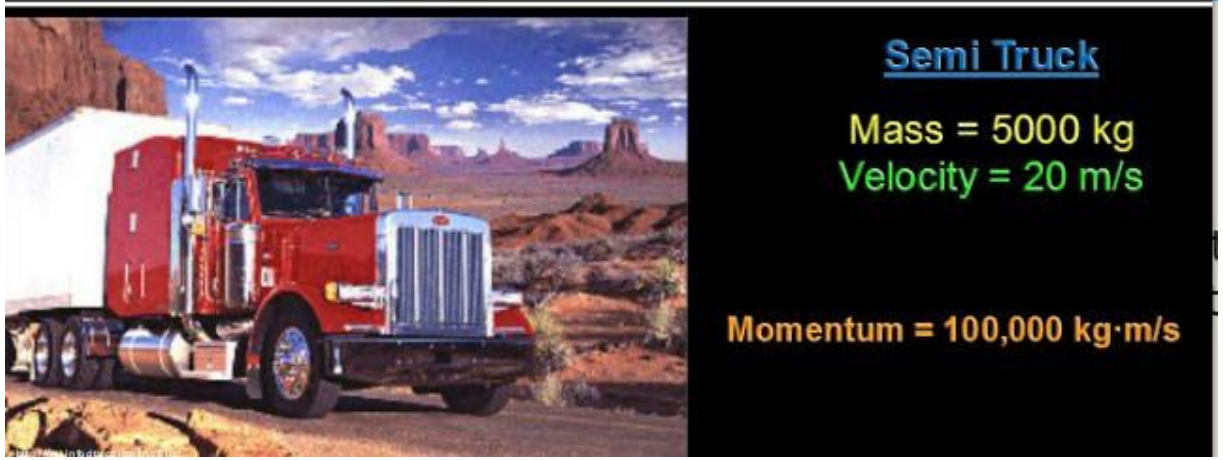
UNITS

Momentum = kg · m/s

$$\text{Momentum} = \frac{\text{kg} \cdot \text{m}}{\text{s}}$$

Momentum

Tomahawk Cruise Missile
Mass = 1000 kg
Velocity = 100 m/s
Momentum = 100,000 kg·m/s


Semi Truck
Mass = 5000 kg
Velocity = 20 m/s
Momentum = 100,000 kg·m/s

Momentum and Newton's laws of motion

- **Newton's first law of motion states that: An object remains at rest or in uniform motion unless acted on by a force.**
- **This law tells us that force is needed to change the momentum of an object.**
- **If the momentum of an object is constant, there is no**
- **resultant force acting on it.**
- **This means that if the mass of an object is constant and the object has a constant momentum, it follows that the velocity of the object is also constant.**

Momentum and Newton's laws of motion

- **Newton's second law of motion states that; the rate of change of momentum of an object is proportional to the resultant force on it. In other words, the resultant force is proportional to the change of momentum per second.**
- **Consider an object of mass *m* acted on by a constant force *F*. Its acceleration causes a change of its speed from initial speed *u* to speed *v* in time *t* without change of direction.**
  - **Its initial momentum = *mu* and its final momentum = *mv***
  - **Its change of momentum = Its final momentum (*mv*) – its initial momentum (*mu*)**

Momentum and Newton's laws of motion

- **According to Newton's second law, the force is directly proportional to the change of momentum per second.**
- **Therefore,**

$$\text{Resultant force} \propto \frac{\text{C}\quad\text{hange in momentum}}{\text{time taken}}$$

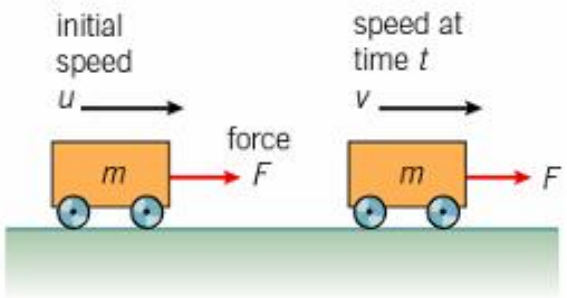
$$F = \frac{\Delta P}{\Delta t}$$

Resultant force = rate of change of momentum

Momentum and Newton's laws of motion

$$F = \frac{\Delta p}{\Delta t} = \frac{mv - mu}{t}$$

$$F = \frac{m(v-u)}{t} = m\left(\frac{v-u}{t}\right) = ma$$



▲ **Figure 2** *Force and momentum*

Where $a = \frac{(v-u)}{t}$ is the acceleration of the object.

4.2 Change in Momentum and Impulse

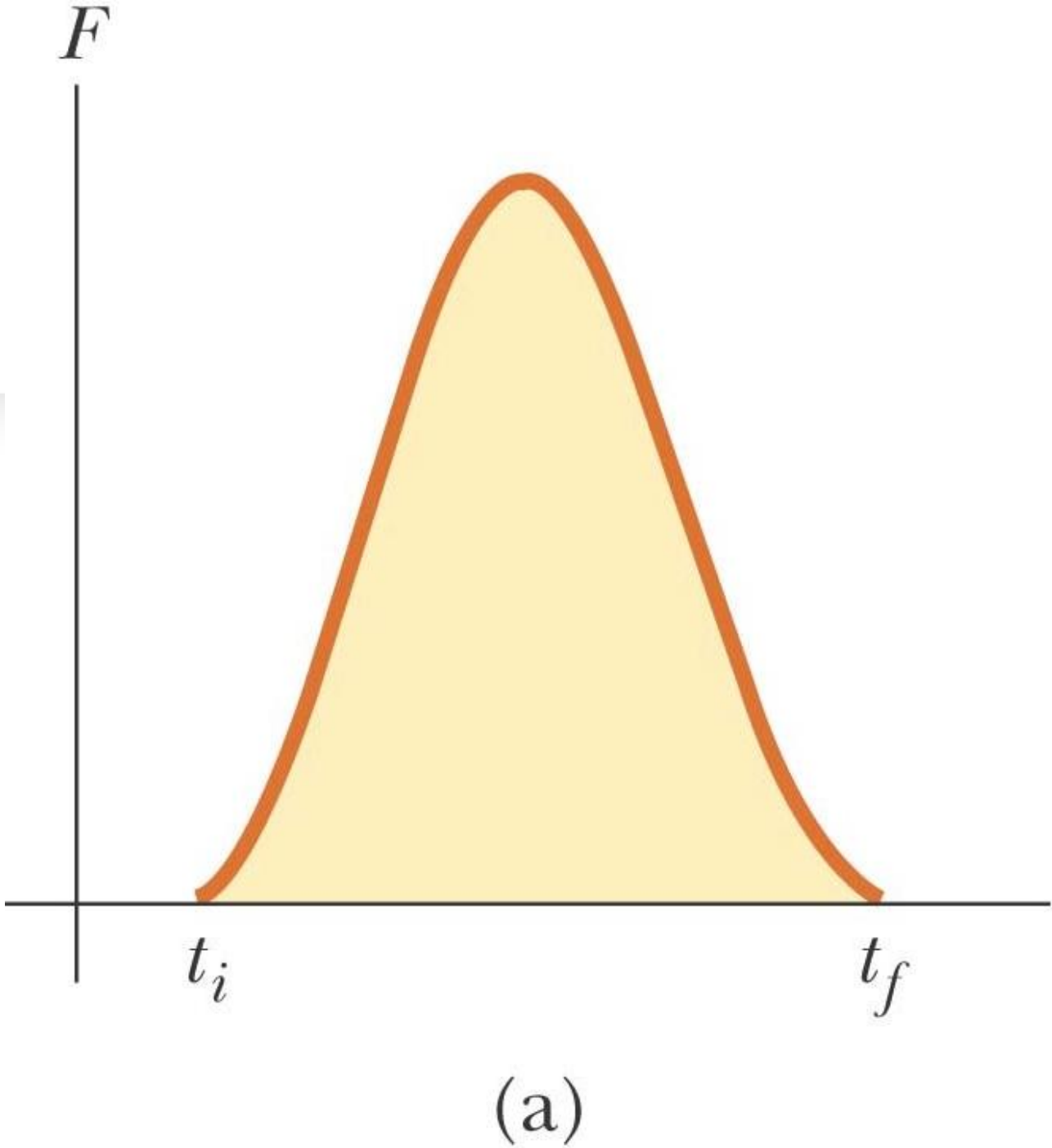
- **The impulse of a force is defined as force x the time for which the force acts. Therefore, for a force F which acts for time Δt,**

$$\textbf{the impulse} = F\Delta t = \Delta(mv)$$

Hence the impulse of a force acting on an object is equal to the change of momentum of the object.

Impulse is a vector quantity with the same direction as the direction of the change in momentum.

- **Impulse has a magnitude equal to the area under the force-time graph.**



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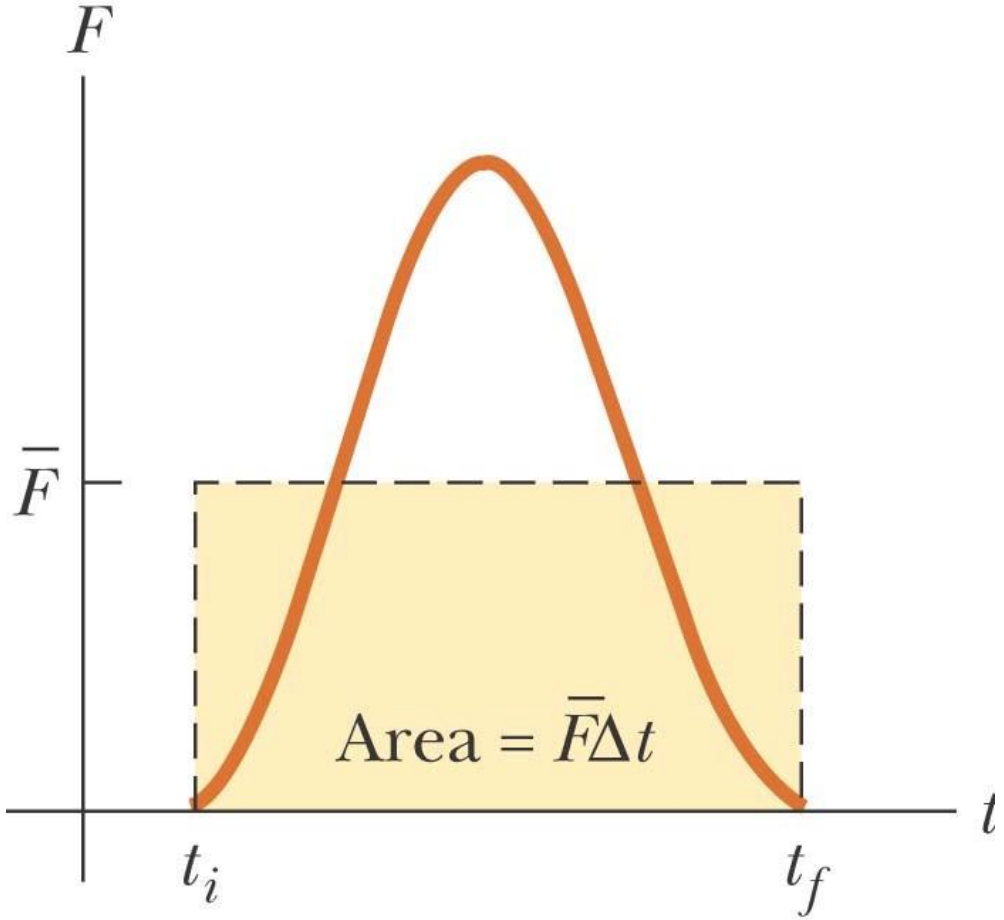
Because the force imparting an impulse can generally vary in time, we can express impulse as

$$\boxed{I = \bar{F}\Delta t}$$



The harder a ball is hit, the further it travels. The impact changes the momentum of the ball in a very short time when the object exerting the impact force is in contact with the ball.

- **If the ball is initially stationary and impact causes it to accelerate to speed v in time t, the gain of momentum of the ball due to impact = mv, where m is the mass of the ball.**

Therefore,

$$the\ force\ of\ impact\ F = \frac{change\ in\ momentum}{contact\ time} = \frac{mv}{t}$$

- **If the ball is moving with an initial velocity u, and the impact changes its velocity to v in time t, the change in momentum of the ball = mv – mu.**

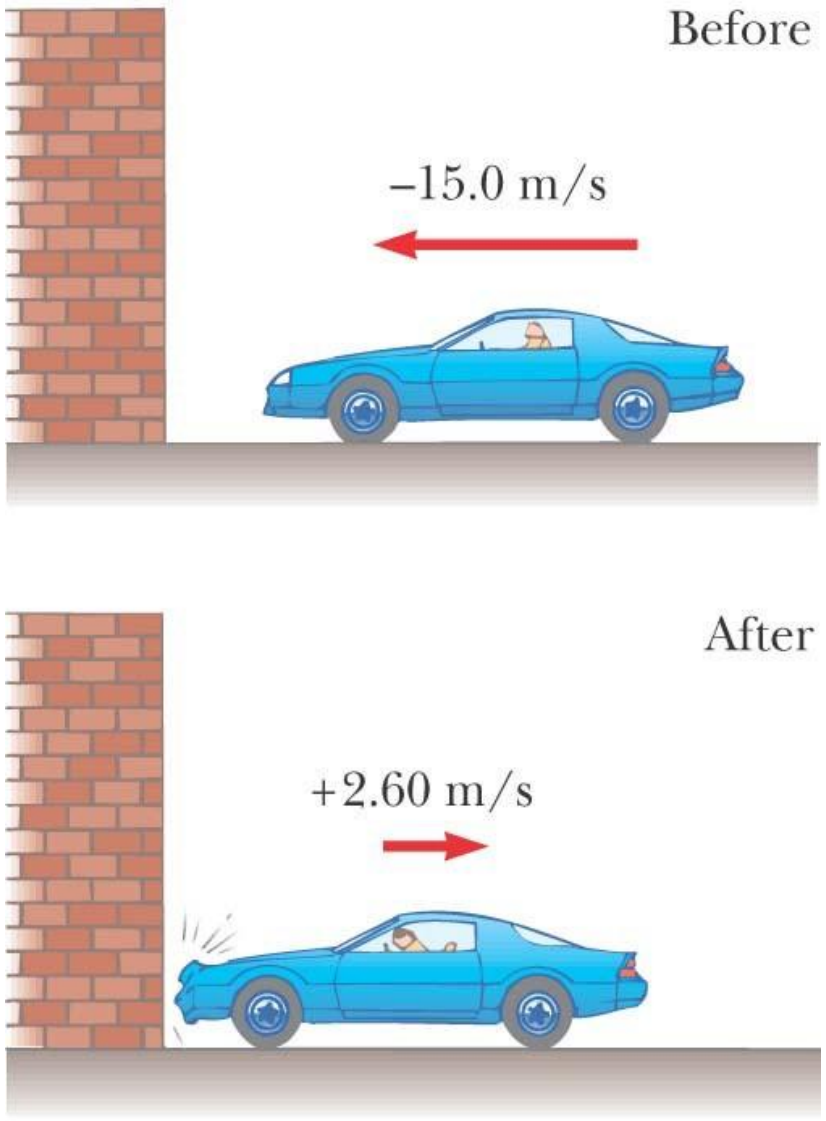
Therefore,

$$the\ force\ of\ impact\ F = \frac{change\ in\ momentum}{contact\ time}$$

$$= \frac{mv - mu}{t}$$

Momentum and Impulse: Example 9.3

In a particular crash test, a car of mass 1500 kg collides with a wall, as shown. The initial and final velocities of the car are vi = -15.0i m/s and vf = 2.6^0i m/s, respectively.



(a)

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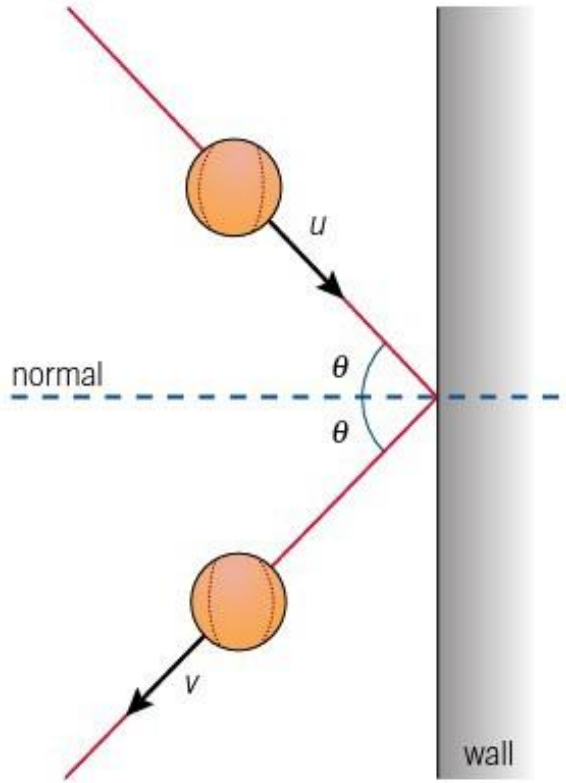
a) If the collision lasts for 0.150 s, find the impulse caused by the collision and the average force exerted on the car.

Oblique Rebound impacts

2. If the impact force is oblique (i.e., the initial direction of the ball is not perpendicular to the wall as shown in figure 5), the normal components of velocity must be used.



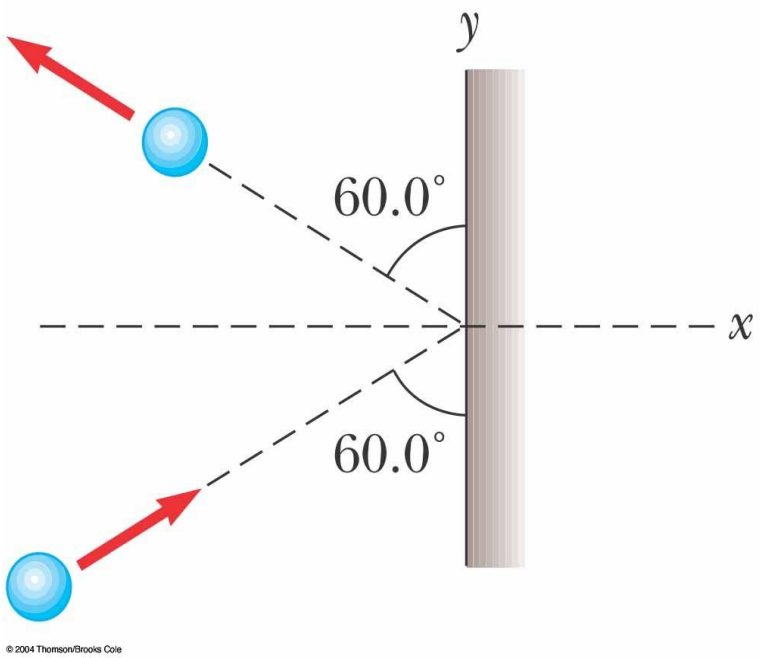
▲ **Figure 5** *An oblique impact*

Oblique Rebound impacts

- **For an impact in which the initial and final direction of the ball are at the same angle θ to the normal and there is no loss of speed (i.e. u = v), the normal component of initial velocity is + u cos θ and the normal component of the final velocity is – u cos θ.**
- **The change of momentum of the ball is therefore**

- 2mu cos θ.

Example 2



A 3.00 kg steel ball strikes a wall with a speed of 10.0 m/s at an angle of 60.0 degree with the surface. It bounces off with the same speed and angle. If the ball is in contact with the wall for 0.200 s, what is the average force exerted by the wall on the ball ?

4.3 Conservation of Linear Momentum

Whenever two or more particles in an isolated system interact, the total momentum of the system remains constant.

The total momentum of an isolated system at all times equals its initial momentum.

$$\boxed{\text{Total p before} = \text{Total p after}}$$

Conservation of Linear Momentum

Whenever two or more particles in an isolated system interact, the total momentum of the system remains constant.

The total momentum of an isolated system at all times equals its initial momentum.

$$\boxed{\text{Total p before} = \text{Total p after}}$$

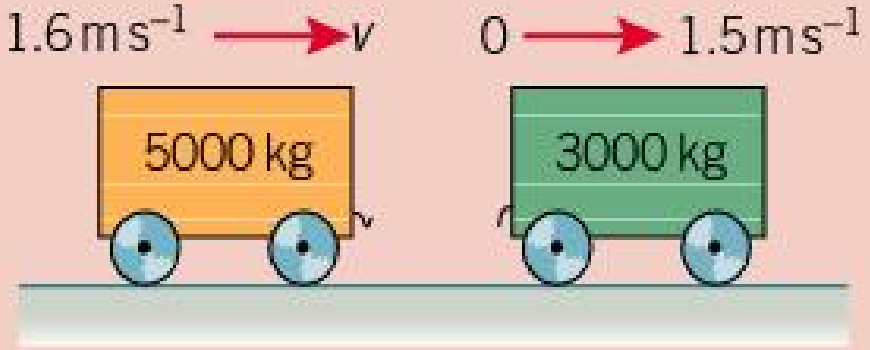
Momentum of a system of particles

$$\boxed{P = \sum_i m_i v_i} = M v_{cm}$$

Which also yields the formula to calculate the velocity of the center of mass:

$$\boxed{v_{cm} = \frac{\sum m_i v_i}{M}}$$

2 A rail wagon of mass 5000 kg moving at a velocity of $1.6\,m\,s^{-1}$ collides with a stationary wagon of mass 3000 kg (see Figure 4). After the collision, the 3000 kg wagon moves away at a velocity of $1.5\,m\,s^{-1}$. Calculate the speed and direction of the 5000 kg wagon after the collision.



Example 9.5 Carry Collision Insurance! AM

An 1 800-kg car stopped at a traffic light is struck from the rear by a 900-kg car. The two cars become entangled, moving along the same path as that of the originally moving car. If the smaller car were moving at 20.0 m/s before the collision, what is the velocity of the entangled cars after the collision?

Conservation of Linear Momentum : Example 9.1

A 60. kg archer stands at rest on frictionless ice and fires a 0.50 kg arrow horizontally at 50. m/s.

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a) With what velocity does the archer move across the ice after firing the arrow.

SOLUTION

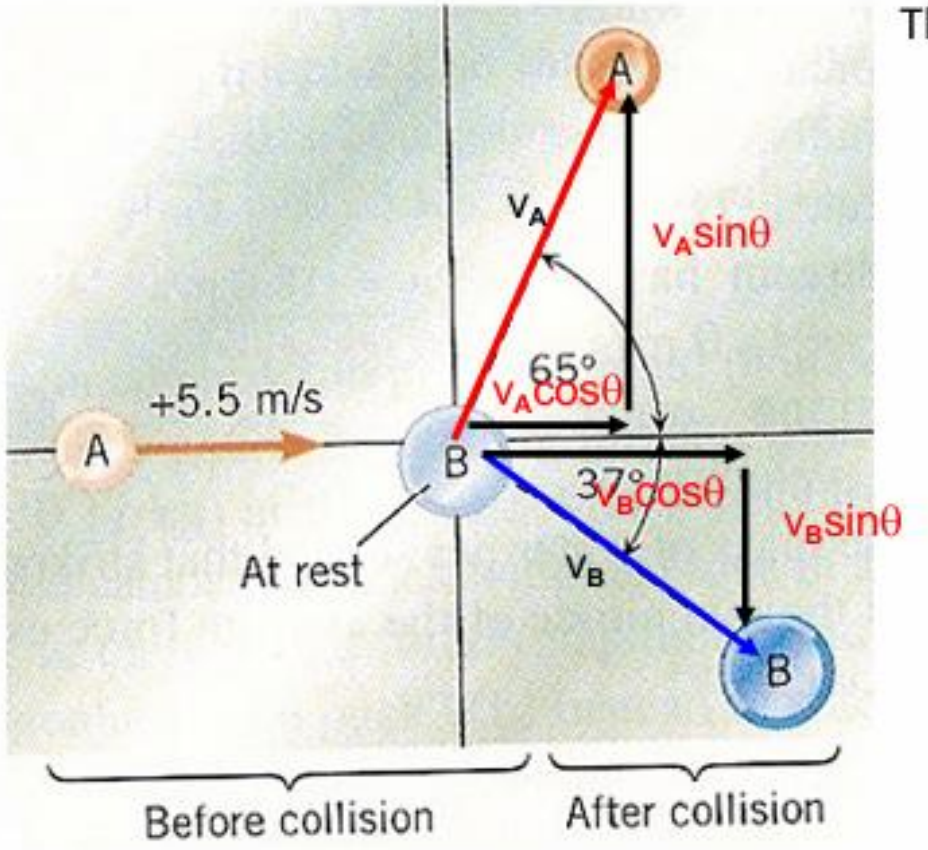
$$p_{ix} = p_{fx}$$

$$0 = m_1 v_{1f} + m_2 v_{2f}$$

$$v_{1f} = -\frac{m_2}{m_1} v_{2f} = -\left(\frac{0.030\ 0\ \text{kg}}{59.97\ \text{kg}}\right)(50.0\ \text{m/s})$$

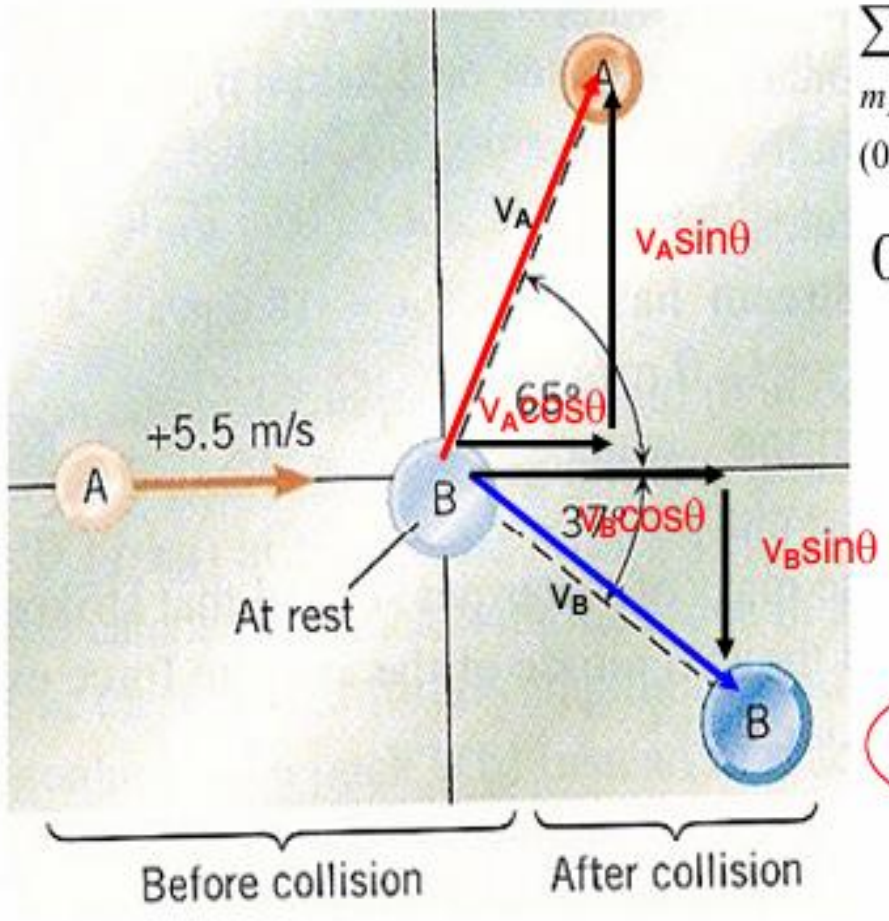
$$v_{1f} = \boxed{-0.025\ 0\ \text{m/s}}$$

Collisions in 2 Dimensions



The figure to the left shows a collision between two pucks on an air hockey table. Puck A has a mass of 0.025-kg and is moving along the x-axis with a velocity of +5.5 m/s. It makes a collision with puck B, which has a mass of 0.050-kg and is initially at rest. The collision is NOT head on. After the collision, the two pucks fly apart with angles shown in the drawing. Calculate the speeds of the pucks after the collision.

Collisions in 2 dimensions



$$\sum p_{ox} = \sum p_x$$

$$m_A v_{oxA} + m_B v_{oxB} = m_A v_{xA} + m_B v_{xB}$$

$$(0.025)(5.5) + 0 = (.025)(v_A \cos 65) + (.050)(v_B \cos 37)$$

$$0.1375 = 0.0106 v_A + 0.040 v_B$$

$$\sum p_{oy} = \sum p_y$$

$$0 = m_A v_{yA} + m_B v_{yB}$$

$$0 = (0.025)(v_A \sin 65) + (0.050)(-v_B \sin 37)$$

$$0.0300 v_B = 0.0227 v_A$$

$$v_B = 0.757 v_A$$

Collisions in 2 dimensions

$$0.1375 = 0.0106v_A + 0.040v_B$$

$$v_B = 0.757v_A$$

$$0.1375 = 0.0106v_A + (0.040)(0.757v_A)$$

$$0.1375 = 0.0106v_A + 0.03785v_A$$

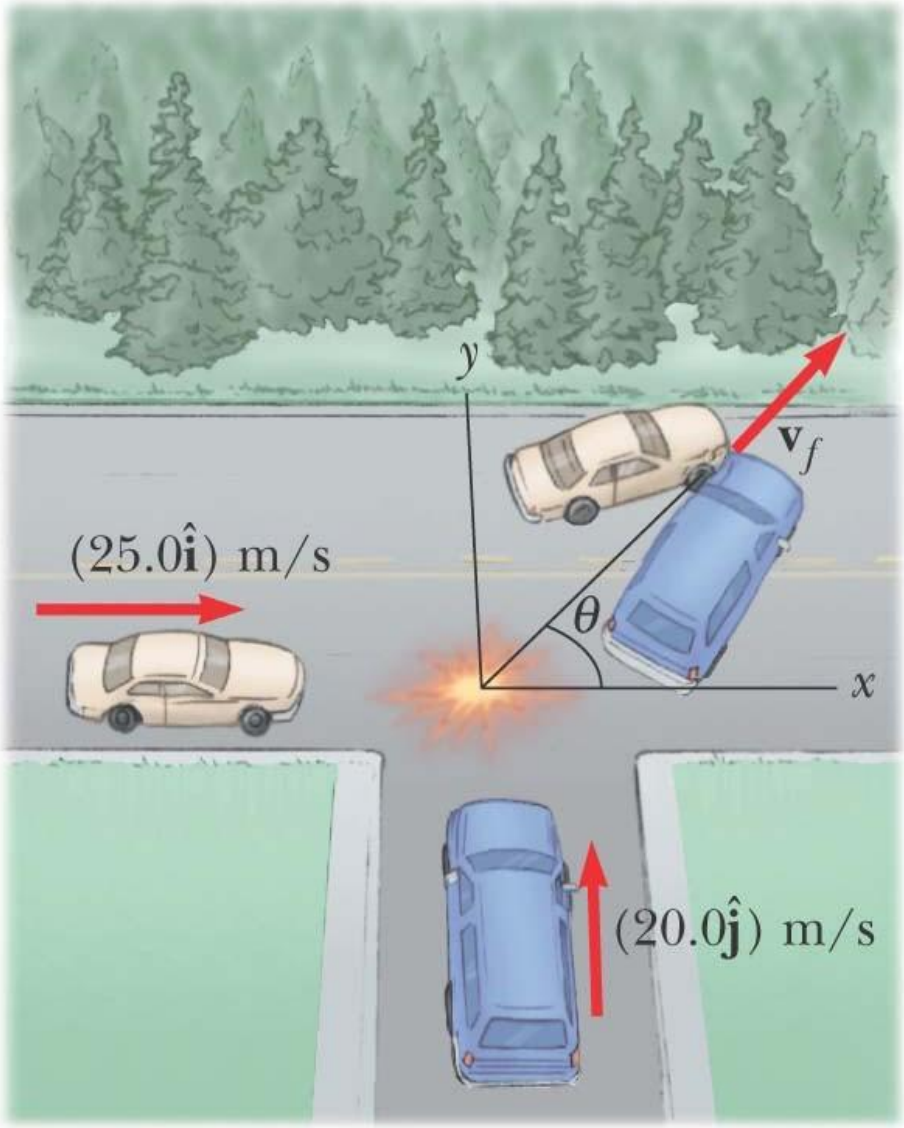
$$0.1375 = 0.04845v_A$$

$$v_A = 2.84m/s$$

$$v_B = 0.757(2.84) = 2.15m/s$$

Conservation of Linear Momentum: Example 9.8

A 1500 kg car traveling east with a speed of 25.0 m/s collides at an intersection with a 2500 kg van traveling north at a speed of 20.0 m/s. Find the direction and magnitude of the velocity of the wreckage after the collision, assuming that the vehicles undergo a perfectly inelastic collision (that is, they stick together).



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SOLUTION

Find the x-components of the initial and final total momenta:

$$\sum p_{xi} = m_{\text{car}} v_{\text{car}} = (1.50 \times 10^3 \text{ kg})(25.0 \text{ m/s})$$
$$= 3.75 \times 10^4 \text{ kg} \cdot \text{m/s}$$
$$\sum p_{xf} = (m_{\text{car}} + m_{\text{truck}}) v_f \cos\theta = (4.00 \times 10^3 \text{ kg}) v_f \cos\theta$$

Set the initial x-momentum equal to the final x-momentum:

$$\textbf{(1)} \quad 3.75 \times 10^4 \text{ kg} \cdot \text{m/s} = (4.00 \times 10^3 \text{ kg}) v_f \cos\theta$$

Find the y-components of the initial and final total momenta:

$$\sum p_{iy} = m_{\text{truck}} v_{\text{truck}} = (2.50 \times 10^3 \text{ kg})(20.0 \text{ m/s})$$
$$= 5.00 \times 10^4 \text{ kg} \cdot \text{m/s}$$
$$\sum p_{fy} = (m_{\text{car}} + m_{\text{truck}}) v_f \sin\theta = (4.00 \times 10^3 \text{ kg}) v_f \sin\theta$$

Divide Equation (2) by Equation (1) and solve for θ:

$$\tan\theta = \frac{5.00 \times 10^4 \text{ kg} \cdot \text{m/s}}{3.75 \times 10^4 \text{ kg} \cdot \text{m}} = 1.33$$
$$\theta = \boxed{53.1^\circ}$$

Substitute this angle back into Equation (2) to find v_f:

$$v_f = \frac{5.00 \times 10^4 \text{ kg} \cdot \text{m/s}}{(4.00 \times 10^3 \text{ kg}) \sin 53.1^\circ} = \boxed{15.6 \text{ m/s}}$$

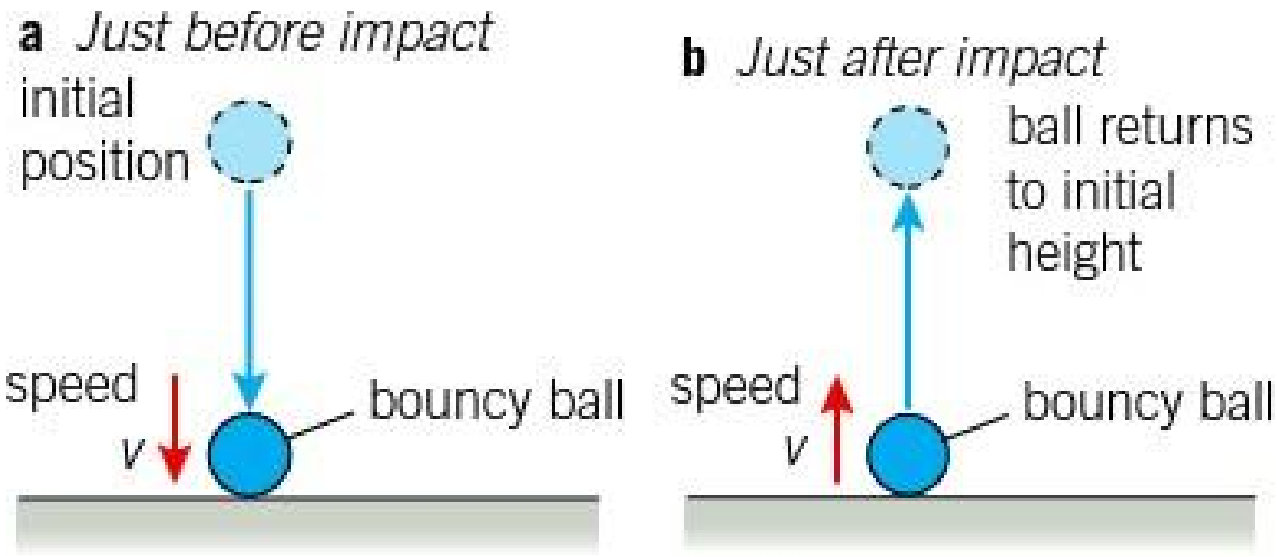
Example 9.9 Proton–Proton Collision AM

A proton collides elastically with another proton that is initially at rest. The incoming proton has an initial speed of 3.50×10^5 m/s and makes a glancing collision with the second proton as in Figure 9.11. (At close separations, the protons exert a repulsive electrostatic force on each other.) After the collision, one proton moves off at an angle of 37.0° to the original direction of motion and the second deflects at an angle of ϕ to the same axis. Find the final speeds of the two protons and the angle ϕ.

4.4 Elastic and Inelastic Collisions

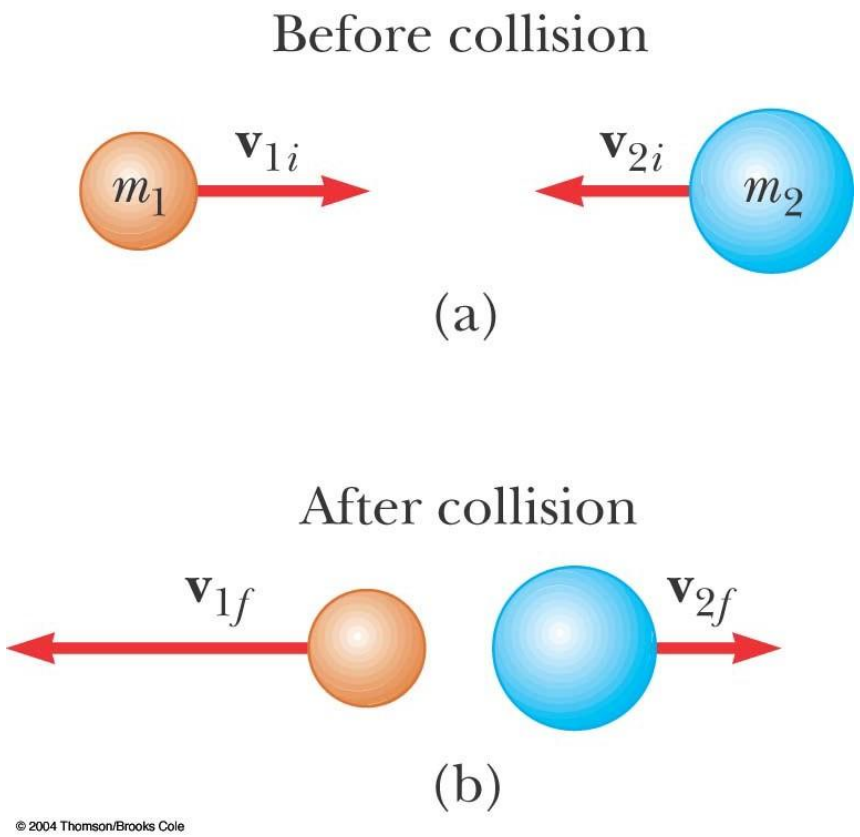
- **An elastic collision is one where there is no loss of kinetic energy.**
- **In this collision, both momentum and kinetic energy are conserved.**
- **For example, when an elastic rubber ball is dropped from a measured height onto a hard floor, the ball will bounce back to exactly the same height.**

Example of perfectly elastic collision



▲ **Figure 1** *An elastic impact*

Elastic Collisions



$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$\tfrac{1}{2} m_1 v_{1i}^2 + \tfrac{1}{2} m_2 v_{2i}^2 = \tfrac{1}{2} m_1 v_{1f}^2 + \tfrac{1}{2} m_2 v_{2f}^2$$

Inelastic collisions

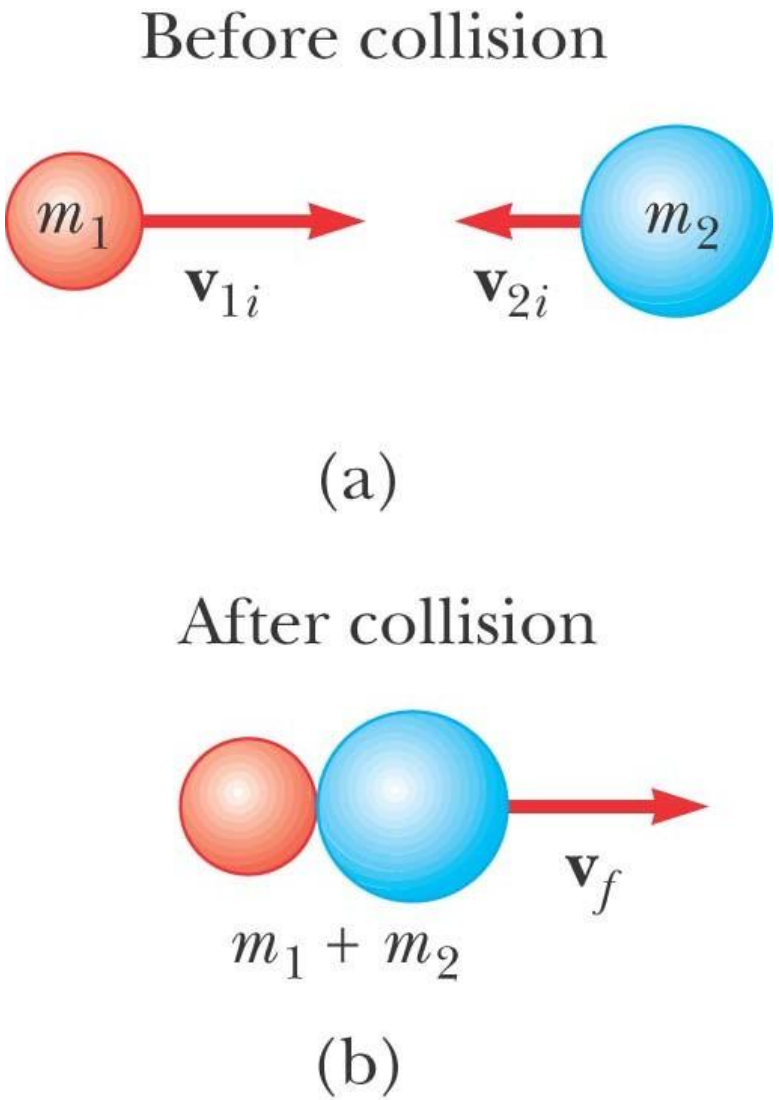
- **In an inelastic collision, the kinetic energy after collision is less than kinetic energy before collision.**

- **Objects that collide and couple together undergo inelastic collision as some of the initial kinetic energy is transferred to the surroundings.**

Perfectly Inelastic Collisions

when the colliding objects stick together

$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$$



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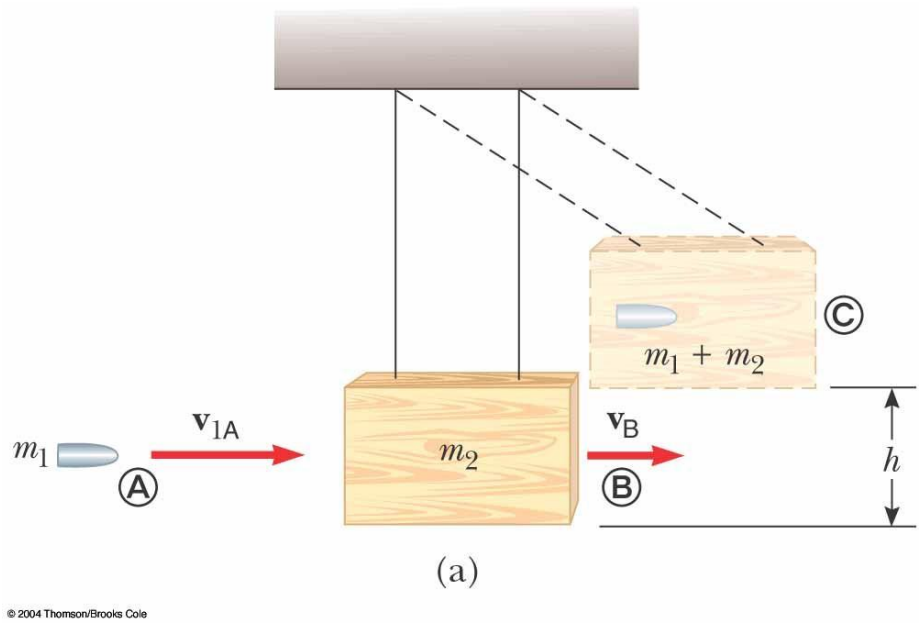
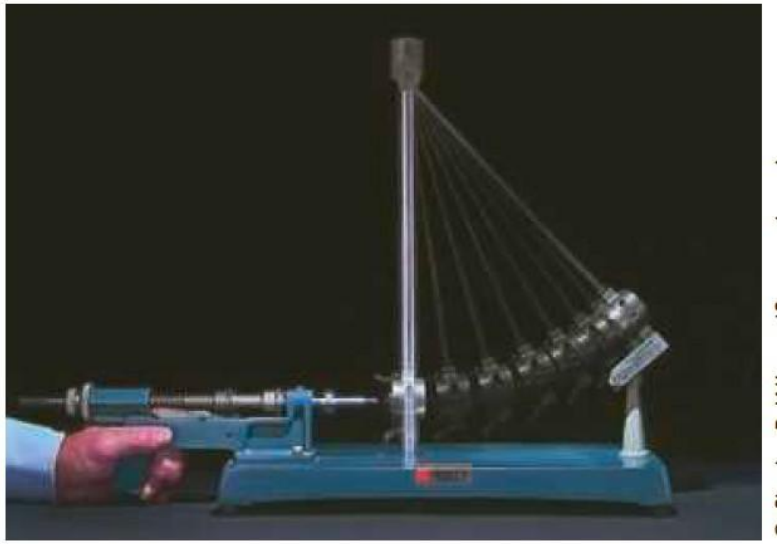
Example 9.8





The ballistic pendulum is an apparatus used to measure the speed of a fast-moving projectile, such as a bullet. A bullet of mass m1 is fired into a large block of wood of mass m2 suspended from some light wires. The bullet embeds in the block, and the entire system swings through a height h. How can we determine the speed of the bullet from a measurement of h ?

Solution

Apply conservation of energy to the block–bullet system after the collision:

$$(KE + PE)_{\text{after collision}} = (KE + PE)_{\text{top}}$$

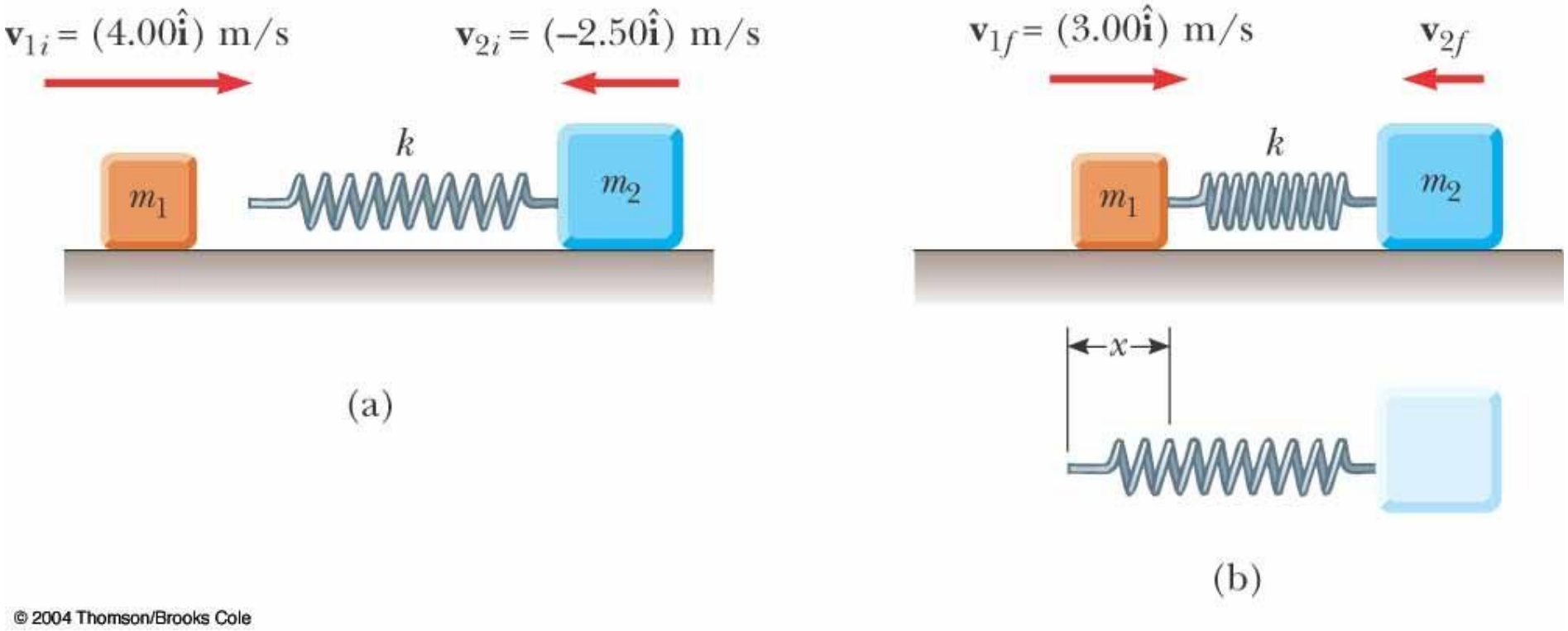
Substitute expressions for the kinetic and potential energies. Note that both the potential energy at the bottom and the kinetic energy at the top are zero:

$$\tfrac{1}{2}(m_1 + m_2)v_{\text{sys}}^2 + 0 = 0 + (m_1 + m_2)gh$$

Solve for the final velocity of the block–bullet system, v_{sys}:

$$v_{\text{sys}}^2 = 2gh$$

Example 3



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A block of mass m1 = 1.60 kg initially moving to the right with a speed of 4.00 m/s on a frictionless horizontal track collides with a spring attached to a second block of mass m2 = 2.10 kg initially moving to the left with a speed of 2.50 m/s. The spring constant is 600 N/m.

a) Find the velocities of the two blocks after the collision.

$$(1) \quad m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$v_{2f} = \frac{m_1 v_{1i} + m_2 v_{2i} - m_1 v_{1f}}{m_2}$$

$$= \frac{(1.60\ \text{kg})(4.00\ \text{m/s}) + (2.10\ \text{kg})(-2.50\ \text{m/s}) - (1.60\ \text{kg})(3.00\ \text{m/s})}{2.10\ \text{kg}}$$

$$v_{2f} = \boxed{-1.74\ \text{m/s}}$$

b) During the collision, at the instant block 1 is moving to the right with a velocity of +3.00 m/s, determine the velocity of block 2.

(c) Find the compression of the spring at that time.

Use energy conservation for the system, noticing that potential energy is stored in the spring when it is compressed a distance x:

$$E_i = E_f$$

$$\frac{1}{2}m_1 v_{1i}^2 + \frac{1}{2}m_2 v_{2i}^2 + 0 = \frac{1}{2}m_1 v_{1f}^2 + \frac{1}{2}m_2\, v_{2f}^2 + \frac{1}{2}kx^2$$

$x =$ 0.173 m

Explosions

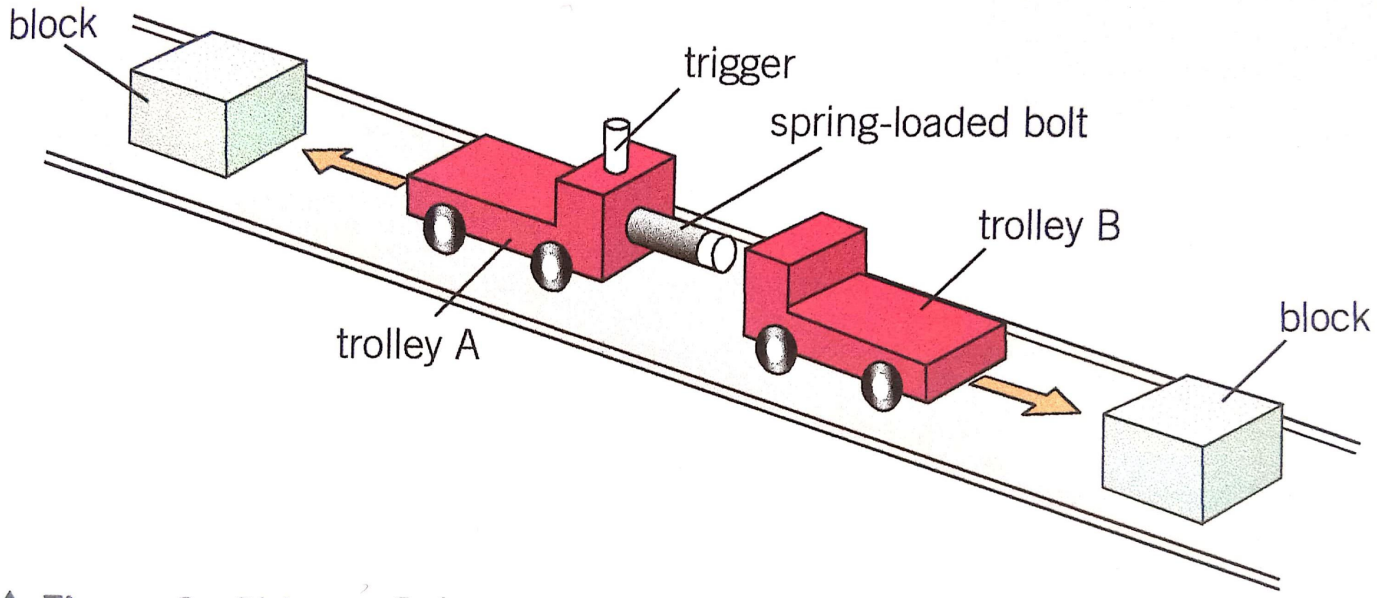
- **When two objects fly apart after being initially at rest, they recoil from each other with equal and opposite amounts of momentum. So they move away from each other in opposite directions.**
- **Consider figure 2 where two spring-loaded trolleys of mass**

  m_A and m_B respectively are initially positioned at rest and in contact. These trolleys move apart at speeds v_A and v_B respectively when the trigger is trapped to release the spring in trolley A.

Explosions

▲ **Figure 1** *The gun barrel recoils when the shell is fired. Large springs fitted to the barrel take away and store the kinetic energy of the barrel as it recoils.*

Explosions



▲ **Figure 2** *Objects flying apart*

Explosions

The total initial momentum is = 0

The total momentum immediately after the explosion

= momentum of A + momentum of B

= $m_A v_A + m_B v_B$

Using the principle of conservation of momentum, $m_A v_A + m_B v_V = 0$

$$\therefore\ m_B v_B = -m_A v_A$$

The minus sign means that the two masses move apart from each other in opposite directions.

Example 9.14 The Exploding Rocket AM

A rocket is fired vertically upward. At the instant it reaches an altitude of 1 000 m and a speed of $v_i = 300$ m/s, it explodes into three fragments having equal mass. One fragment moves upward with a speed of $v_1 = 450$ m/s following the explosion. The second fragment has a speed of $v_2 = 240$ m/s and is moving east right after the explosion. What is the velocity of the third fragment immediately after the explosion?

Example 9.15 Pushing on a Spring[3] AM

As shown in Figure 9.22a (page 276), two blocks are at rest on a frictionless, level table. Both blocks have the same mass m, and they are connected by a spring of negligible mass. The separation distance of the blocks when the spring is relaxed is L. During a time interval Δt, a constant force of magnitude F is applied horizontally to the left block, moving it through a distance x_1 as shown in Figure 9.22b. During this time interval, the right block moves through a distance x_2. At the end of this time interval, the force F is removed.

(A) Find the resulting speed $\vec{v}_{CM}$ of the center of mass of the system.

(B) Find the total energy of the system associated with vibration relative to its center of mass after the force F is removed.

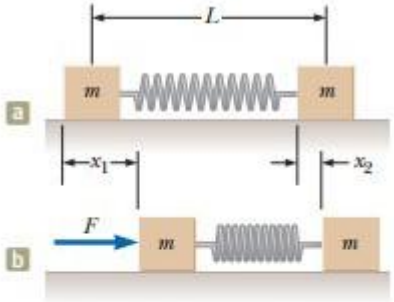


Figure 9.22 (Example 9.15) (a) Two blocks of equal mass are connected by a spring. (b) The left block is pushed with a constant force of magnitude F and moves a distance x_1 during some time interval. During this same time interval, the right block moves through a distance x_2.

Example 9.16 Fighting a Fire

Two firefighters must apply a total force of 600 N to steady a hose that is discharging water at the rate of 3 600 L/min. Estimate the speed of the water as it exits the nozzle.

Example 9.17 A Rocket in Space

A rocket moving in space, far from all other objects, has a speed of 3.0×10^3 m/s relative to the Earth. Its engines are turned on, and fuel is ejected in a direction opposite the rocket's motion at a speed of 5.0×10^3 m/s relative to the rocket.

(A) What is the speed of the rocket relative to the Earth once the rocket's mass is reduced to half its mass before ignition?

(B) What is the thrust on the rocket if it burns fuel at the rate of 50 kg/s?

Example Question of AP Classroom

A pendulum has a length l and a bob of mass m. Which of the following is true of the linear momentum of the bob as it swings from its highest to lowest point?

(A) It remains constant because momentum is always conserved.

(B) It increases in magnitude and changes direction.

(C) It decreases in magnitude and does not change direction.

(D) It is converted to angular momentum.

(E) It is converted to kinetic energy.

Example Question of AP Classroom

A 12 kg box sliding on a horizontal floor has an initial speed of 4.0 m/s. The coefficient of friction between the box and the floor is 0.20. The box moves a distance of 4.0 m in 2.0 s. The magnitude of the change in momentum of the box during this time is most nearly

(A) 12 kg·m/s

(B) 48 kg·m/s

(C) 60 kg·m/s

(D) 96 kg·m/s

Example Question of AP Classroom

A dog of mass 10 kg sits on a skateboard of mass 2 kg that is initially traveling south at 2 m/s. The dog jumps off with a velocity of 1 m/s north rela-tive to the ground. Which of the following is the best estimate of the velocity of the skateboard immediately after the dog has jumped?

(A) 1 m/s north

(B) 1 m/s south

(C) 3 m/s south

(D) 7 m/s south

(E) 17 m/s south